

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/2024 Semester 1 MH1100 Calculus I

Tutorial 12, Week 13

Your tutor will aim to discuss: Problem 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10

Problem 1 Find a formula for the inverse of the function

(a)

$$f(x) = \frac{4x - 1}{2x + 3}$$

(b)

$$f(x) = \frac{1 - \sqrt{x}}{1 + \sqrt{x}}$$

[Solution:] (a) We have

$$y = f(x) = \frac{4x - 1}{2x + 3} \Rightarrow y(2x + 3) = 4x - 1 \Rightarrow 2xy + 3y = 4x - 1 \Rightarrow 3y + 1 = 4x - 2xy$$

$$\Rightarrow 3y + 1 = (4 - 2y)x \Rightarrow x = \frac{3y + 1}{4 - 2y}$$

We then interchange x and y , and have

$$y = f^{-1}(x) = \frac{3x + 1}{4 - 2x}$$

(b) We have

$$y = \frac{1 - \sqrt{x}}{1 + \sqrt{x}} \Rightarrow y(1 + \sqrt{x}) = 1 - \sqrt{x} \Rightarrow \sqrt{x}(1 + y) = 1 - y$$

$$\Rightarrow \sqrt{x} = \frac{1 - y}{1 + y} \Rightarrow x = \left(\frac{1 - y}{1 + y}\right)^2$$

We then interchange x and y , and have

$$y = f^{-1}(x) = \left(\frac{1 - x}{1 + x}\right)^2$$

Problem 2 If $h(x) = x + \sqrt{x}$, find $h^{-1}(6)$.

[Solution:] We have

$$h'(x) = 1 + \frac{1}{2\sqrt{x}} > 0$$

Therefore, the original function is a increasing function in its domain $[0, \infty)$. Hence it is one to one.

Further we have $h(4) = 4 + \sqrt{4} = 6$, so $h^{-1}(6) = 4$.

Problem 3 Find $(f^{-1})'(a)$ for

(a)

$$f(x) = 3 + x^2 + \tan\left(\frac{\pi x}{2}\right), \quad -1 < x < 1, a = 3$$

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(b)

$$f(x) = \sqrt{x^3 + 4x + 4}, \quad a = 3$$

[Solution:] (a) We have $f(0) = 3 \Rightarrow f^{-1}(3) = 0$. Further

$$f(x) = 3 + x^2 + \tan\left(\frac{\pi x}{2}\right) \Rightarrow f'(x) = 2x + \frac{\pi}{2} \sec^2\left(\frac{\pi x}{2}\right)$$

and $f'(0) = \frac{\pi}{2} \cdot 1 = \frac{\pi}{2}$. Thus,

$$(f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(0)} = \frac{2}{\pi}$$

(b) We have $f(1) = 3 \Rightarrow f^{-1}(3) = 1$. Further

$$f(x) = \sqrt{x^3 + 4x + 4} \Rightarrow f'(x) = \frac{3x^2 + 4}{2\sqrt{x^3 + 4x + 4}}$$

and $f'(1) = \frac{7}{6}$. Thus

$$(f^{-1})'(3) = \frac{1}{f'(f^{-1}(3))} = \frac{1}{f'(1)} = \frac{6}{7}$$

Problem 4 Find the limit

(a)

$$\lim_{x \rightarrow \infty} (1.001)^x$$

(b)

$$\lim_{x \rightarrow -\infty} (1.001)^x$$

[Solution:] We know when $b > 1$, then $\lim_{x \rightarrow \infty} b^x = \infty$ and $\lim_{x \rightarrow -\infty} b^x = 0$ So we have

$$(a) \quad \lim_{x \rightarrow \infty} (1.001)^x = \infty$$

and

$$(b) \quad \lim_{x \rightarrow -\infty} (1.001)^x = 0$$

Problem 5 Differentiate the function

(a)

$$y = e^{ax^3}$$

(b)

$$y = \ln \ln \ln(x)$$

(c)

$$y = \log_2(x \log_5(x))$$

[Solution:] (a) We have

$$y' = e^{ax^3} (ax^3)' = 3ax^2 e^{ax^3}.$$

(b) We have

$$y' = \frac{(\ln \ln(x))'}{\ln \ln(x)} = \frac{\frac{1/x}{\ln(x)}}{\ln \ln(x)} = \frac{1}{(\ln \ln(x))(\ln x)x}$$

(c) We have

$$y' = \frac{1}{\ln 2} \frac{(x \log_5(x))'}{x \log_5(x)} = \frac{1}{\ln 2} \frac{\log_5(x) + x \frac{1/x}{\ln 5}}{x \log_5(x)} = \frac{1}{\ln 2} \frac{\log_5(x) + \frac{1}{\ln 5}}{x \log_5(x)}$$

$$= \frac{1}{\ln 2} \left(\frac{1}{x} + \frac{1}{x \log_5(x) \ln 5} \right) = \frac{1}{\ln 2} \left(\frac{1}{x} + \frac{1}{x \ln(x)} \right)$$

Problem 6 Use logarithm differentiation to find the derivative of the function.

(a)

$$y = \sqrt{x} e^{x^2-x} (x+1)^{2/3}$$

(b)

$$y = (\sin x)^{\ln x}$$

[Solution:] (a) We have

$$\begin{aligned} \ln(y) &= \frac{1}{2} \ln x + x^2 - x + \frac{2}{3} \ln(x+1) \Rightarrow \frac{y'}{y} = \frac{1}{2x} + 2x - 1 + \frac{2}{3(x+1)} \\ \Rightarrow y' &= (\sqrt{x} e^{x^2-x} (x+1)^{2/3}) \left(\frac{1}{2x} + 2x - 1 + \frac{2}{3(x+1)} \right) \end{aligned}$$

(b) We have

$$\begin{aligned} \ln(y) &= \ln(x)(\ln(\sin x)) \Rightarrow \frac{y'}{y} = \frac{1}{x} (\ln(\sin x)) + \ln(x) \frac{\cos x}{\sin x} \\ \Rightarrow ((\sin x)^{\ln x}) &\left(\frac{1}{x} (\ln(\sin x)) + \ln(x) \frac{\cos x}{\sin x} \right) \end{aligned}$$

Problem 7 Find y' if $x^y = y^x$.

[Solution:] Take the logarithm on both sides

$$x^y = y^x \Rightarrow y \ln x = x \ln y$$

Take the derivative with respect to x ,

$$y' \ln x + \frac{y}{x} = \ln y + \frac{xy'}{y} \Rightarrow y'(\ln x - \frac{x}{y}) = \ln y - \frac{y}{x} \Rightarrow y' = \frac{\ln y - \frac{y}{x}}{\ln x - \frac{x}{y}}$$

Problem 8 Solve the equation for x

(a)

$$e^{e^x} = 10$$

(b)

$$\begin{vmatrix} \ln(2x+1) & 2 - \ln x \\ 1 & 1 \end{vmatrix} = 0$$

[Solution:] (a) We have

$$e^{e^x} = 10 \Rightarrow e^x = \ln(10) \Rightarrow x = \ln(\ln(10))$$

(b) We have

$$\begin{aligned} \begin{vmatrix} \ln(2x+1) & 2 - \ln x \\ 1 & 1 \end{vmatrix} &= 0 \Rightarrow \ln(2x+1) - (2 - \ln x) = 0 \\ \Rightarrow \ln(2x+1) + \ln x &= 2 \Rightarrow \ln((2x+1)x) = 2 \Rightarrow 2x^2 + x - e^2 = 0 \end{aligned}$$

Therefore, the root is

$$x = \frac{-1 + \sqrt{1 + 8e^2}}{4}$$

Note that we omit the negative value, as x has to be positive.

Problem 9 Use the definition of derivative to prove that

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1.$$

[Solution:] We let $f(x) = \ln(1+x)$, then $f'(x) = \frac{1}{1+x}$, so $f'(0) = 1$.

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0) = 1$$

Problem 10 Show that for any $x > 0$, we have

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x.$$

[Solution:] We can let $m = n/x$. Note that this is fine as $x > 0$. And $m \rightarrow \infty$, when $n \rightarrow \infty$. Therefore, we have

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n/x}\right)^n = \lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^{mx} = \left(\lim_{m \rightarrow \infty} \left(1 + \frac{1}{m}\right)^m\right)^x = e^x$$